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**Final Project Abstract**



# Abstract: Risk-aware Weighted Buffered Voronoi Tessellation for Decentralized Multi-Agent Collision-Free Navigation of Dynamic Robotic Arms

In the paper authored by Lyu et al. [1], the authors present an innovative approach known as Risk-aware Weighted Buffered (RAWB) Voronoi Cell, a variant of Generalized Voronoi tessellation. This approach is designed to create safe planning sets for decentralized multi-agent collision-free navigation. The primary objective of their research is to ensure inter-robot collision avoidance of dynamic systems by partitioning the joint state space of a multi-agent robotic system into individual cells, thus enabling each agent to navigate in a distributed manner within its own cell while accounting for varying risk levels.

In my project I implemented these RAWB Voroni Cells for dynamic 2D robotic arms. I made a 2D simulation of dynamic robot arms which used RNE dynamics for each step, and could visualize the robots. I then followed a 5 step approach to implementation of the control of each arm where I first: 1) Computed the safe region for each joint with the RAWB Voronoi cells, 2) projected the goal position onto closest point in the safe region of the end effector joint, 3) solved inverse kinematics to reach the projected goal, 4) calculated an input torque to the joints using a poorly tuned computed torque controller, 5) then limited the control input to be within the admissible control space for the RAWB Voronoi Cells as defined in [1] in between Equation 6 and 7. I intentionally used dynamic robots with poorly tuned controllers to illustrate the collision avoidance of the RAWB Voronoi Cells, as opposed to a good controller. This created a simulation in which I could randomize the initial positions and goal positions of the robots, and run multiple trials to validate performance of the model. For comparison I also implemented the Risk-Aware cells as outlined in the paper from [1] which does not provide the same safety guarantees for dynamic simulations.

In trials it was observed that when using RAWB Voronoi cells robots never collided in all simulations, even if this meant the final goal could not be reached due to deadlock conditions. The RA Voronoi cells did not perform safely in all situations indicating that the RAWB Cells provide an advantage over other Voronoi cells. This approach is effective, but also has limitation. Future work to expand on these RAWB cells and overcome shortcomings could include generalizing the approach to 3 dimensions, adding safety checks for link collision as opposed to only joint collisions, and additional validation with more simulations.

(a) 3 robots

(b) 6 robots

Figure 1: Two static visualizations of robot arms operating with Risk-Aware Weighted Buffered Voronoi cells. A Link to additional videos included here.

## References

[1] Y. Lyu, J. M. Dolan, and W. Luo, “Decentralized safe navigation for multi-agent systems via risk-aware weighted buffered voronoi cells,” in *Proceedings of the 2023 International Conference on Autonomous Agents and Multiagent Systems*, ser. AAMAS ’23, London, United Kingdom: International Foundation for Autonomous Agents and Multiagent Systems, 2023, pp. 1476–1484.